JIAYU LI

+852-5531-1950 | jyli93@hku.hk

LinkedIn | GitHub | Google Scholar

Pokfulam Road, The University of Hong Kong, Hong Kong SAR

Experience

- **The University of Hong Kong** — *Jul. 2024 to present*
  Postdoctoral Fellow — Hong Kong SAR, China
- **The University of Hong Kong** — *May 2024 – Jun. 2024*
  Research Assistant — Hong Kong SAR, China
- **Southern University of Science and Technology** — *Jul. 31, 2019 – Jul. 2020*
  Research Assistant — Shenzhen, Guangdong, China

Education

- **Southern University of Science and Technology** — *Sep. 2020 – Jun. 2024*
  Ph.D. in Condensed Matter Physics — Shenzhen, Guangdong, China
- **South China Normal University** — *Sep. 2016 – Jun. 2019*
  M.S. in Condensed Matter Physics — Guangzhou, Guangdong, China
- **Changsha University of Science and Technology** — *Sep. 2012 – Jun. 2016*
  B.S. in Physics — Changsha, Hunan, China

Research Interests

Theoretical and computational condensed matter physics:

- Spin space groups, spin groups, and quasisymmetry in quantum materials
- Topological phases and quasiparticles in magnetic systems
- Quantum geometry and nonlinear/spin transport in antiferromagnets

Research Profile

I have co-authored 27 publications, including 26 peer-reviewed papers and one preprint. My first/co-first/corresponding-author works include 2 papers in *Phys. Rev. Lett./Phys. Rev. X*, 3 in *Nat. Commun.*, 1 in *Natl. Sci. Rev.*, and 1 in *Adv. Funct. Mater.*; I have also contributed to collaborative works in *Nature* and other leading journals.

Honors and Awards

- **Outstanding Doctoral Dissertation** — *Jun. 2024*
  Southern University of Science and Technology — Shenzhen, Guangdong, China
- **National Scholarship for Graduate Excellence** — *Oct. 2021*
  Southern University of Science and Technology — Shenzhen, Guangdong, China

Publications

† Co-first author, * Corresponding author

1. **J. Li**, F.-R. Fan, W. Yao*, Quasisymmetry enriched gapless criticality at Chern insulator transitions, *arXiv*: 2601.15011 (2026)

2. H. Cai, Q. Yao, **J. Li***, X. Xuan, W. Guo, Z. Zhang*, A symmetry strategy for large bandgap quantum anomalous Hall insulators, *Adv. Funct. Mater.*, e76694 (2026)

3. H. Zhu†, **J. Li**†, X. Chen, Y. Yu, Q. Liu*, Magnetic geometry induced quantum geometry and nonlinear transports, *Nat. Commun.* **16**, 4882 (2025)

4. A. Zhang†, X. Chen†, **J. Li**, P. Liu, Y. Liu, Q. Liu*, Topological charge quadrupole protected by spin-orbit U(1) quasi-symmetry in antiferromagnet NdBiPt, *Newton* **1**, 100010 (2025)

5. X. Chen, Y. Liu, P. Liu, Y. Yu, J. Ren, **J. Li**, A. Zhang, Q. Liu*, Unconventional magnons in collinear magnets dictated by spin space groups, *Nature* **640**, 349 (2025)

6. L. Liu†, Y. Liu†, **J. Li**, H. Wu, Q. Liu*, Quantum spin Hall effect protected by spin U(1) quasisymmetry, *Phys. Rev. B* **110**, L161104 (2024)

7. X. Chen†, J. Ren†, Y. Zhu†, Y. Yu†, A. Zhang, P. Liu, **J. Li**, Y. Liu, C. Li, Q. Liu*, Enumeration and representation theory of spin space groups, *Phys. Rev. X* **14**, 031038 (2024)

8. L. Liu†, Y. Liu†, **J. Li**, H. Wu*, Q. Liu*, Orbital doublet driven even-spin Chern insulators, *Phys. Rev. B* **110**, 035161 (2024)

9. **J. Li**, A. Zhang, Y. Liu, Q. Liu*, Group theory on quasisymmetry and protected near degeneracy, *Phys. Rev. Lett.* **133**, 026402 (2024)

10. Y.-P. Zhu†, X. Chen†, X.-R. Liu, Y. Liu, P. Liu, H. Zha, G. Qu, C. Hong, **J. Li**, Z. Jiang, X.-M. Ma, Y.-J. Hao, M.-Y. Zhu, W. Liu, M. Zeng, S. Jayaram, M. Lenger, J. Ding, S. Mo, K. Tanaka, M. Arita, Z. Liu, M. Ye, D. Shen, J. Wrachtrup, Y. Huang, R.-H. He, S. Qiao*, Q. Liu*, C. Liu*, Observation of plaid-like spin splitting in a noncoplanar antiferromagnet, *Nature* **626**, 523 (2024)

11. Y. Wan†, **J. Li**†, Q. Liu*, Topological magnetoelectric response in ferromagnetic axion insulators, *Natl. Sci. Rev.* **11**, nwac138 (2024)

12. A. Zhang†, K. Deng†, J. Sheng†, P. Liu†, S. Kumar, K. Shimada, Z. Jiang, Z. Liu, D. Shen, **J. Li**, J. Ren, L. Wang, L. Zhou, Y. Ishikawa, T. Ohhara, Q. Zhang, G. McIntyre, D. Yu, E. Liu, L. Wu*, C. Chen*, Q. Liu*, Chiral Dirac fermion in a collinear antiferromagnet, *Chin. Phys. Lett.* **40**, 126101 (2023)

13. Y. Liu, **J. Li**, Q. Liu*, Chern-insulator phase in antiferromagnets, *Nano Lett.* **23**, 8650 (2023)

14. L. L. Tao*, **J. Li**, Y. Liu, X. Wang, Y. Sui, B. Song, M. Y. Zhuravlev, Q. Liu*, Rashba-like spin splitting around non-time-reversal-invariant momenta, *Phys. Rev. B* **107**, 235138 (2023)

15. W. Chen, M. Gu, **J. Li**, P. Wang, Q. Liu*, Role of hidden spin polarization in nonreciprocal transport of antiferromagnets, *Phys. Rev. Lett.* **129**, 276601 (2022)

16. P. Liu, **J. Li**, J. Han, X. Wan*, Q. Liu*, Spin-group symmetry in magnetic materials with negligible spin-orbit coupling, *Phys. Rev. X* **12**, 021016 (2022)

17. **J. Li**†, Q. Yao†, L. Wu†, Z. Hu, B. Gao, X. Wan*, Q. Liu*, Designing light-element materials with large effective spin-orbit coupling, *Nat. Commun.* **13**, 919 (2022)

18. Z. Hao†, W. Chen†, Y. Wang, **J. Li**, X.-M. Ma, Y.-J. Hao, R. Lu, Z. Shen, Z. Jiang, W. Liu, Q. Jiang, Y. Yang, X. Lei, L. Wang, Y. Fu, L. Zhou, L. Huang, Z. Liu, M. Ye, D. Shen, J. Mei, H. He, C. Liu, K. Deng, C. Liu, Q. Liu*, C. Chen*, Multiple Dirac nodal lines in an in-plane anisotropic semimetal $TaNiTe_5$, *Phys. Rev. B* **104**, 115158 (2021)

19. Q. Yao, **J. Li**, Q. Liu*, Fragile symmetry-protected half metallicity in two-dimensional van der Waals magnets: A case study of monolayer $FeCl_2$, *Phys. Rev. B* **104**, 035108 (2021)

20. M. Gu†, **J. Li**†, H. Sun, Y. Zhao, C. Liu, J. Liu*, H. Lu, Q. Liu*, Spectral signatures of the surface anomalous Hall effect in magnetic axion insulators, *Nat. Commun.* **12**, 3524 (2021)

21. X.-M. Ma†, Y. Zhao†, K. Zhang†, S. Kumar†, R. Lu, **J. Li**, Q. Yao, J. Shao, F. Hou, X. Wu, M. Zeng, Y.-J. Hao, Z. Hao, Y. Wang, X.-R. Liu, H. Shen, H. Sun, J. Mei, K. Miyamoto, T. Okuda, M. Arita, E. F. Schwier, K. Shimada, K. Deng, C. Liu, J. Lin, Y. Zhao, C. Chen*, Q. Liu*, C. Liu*, Realization of a tunable surface Dirac gap in Sb-doped $MnBi_2Te_4$, *Phys. Rev. B* **103**, L121112 (2021)

22. R. Lu†, H. Sun†, S. Kumar†, Y. Wang†, M. Gu, M. Zeng, Y.-J. Hao, **J. Li**, J. Shao, X.-M. Ma, Z. Hao, K. Zhang, W. Mansuer, J. Mei, Y. Zhao, C. Liu, K. Deng, W. Huang, B. Shen, K. Shimada, E. F. Schwier*, C. Liu*, Q. Liu*, C. Chen*, Half-magnetic topological insulator with magnetization-induced Dirac gap at a selected surface, *Phys. Rev. X* **11**, 011039 (2021)

23. X. Wu†, **J. Li**†, X.-M. Ma†, Y. Zhang†, Y. Liu, C.-S. Zhou, J. Shao, Q. Wang, Y.-J. Hao, Y. Feng, E. F. Schwier, S. Kumar, H. Sun, P. Liu, K. Shimada, K. Miyamoto, T. Okuda, K. Wang, M. Xie, C. Chen, Q. Liu*, C. Liu*, Y. Zhao*, Distinct topological surface states on the two terminations of $MnBi_4Te_7$, *Phys. Rev. X* **10**, 031013 (2020)

24. S.-H. Zheng, H.-J. Duan, J.-K. Wang, **J.-Y. Li**, M.-X. Deng*, R.-Q. Wang*, Origin of planar Hall effect on the surface of topological insulators: Tilt of Dirac cone by an in-plane magnetic field, *Phys. Rev. B* **101,** 041408(R) (2020)

25. J.-K. Wang, **J.-Y. Li**, H.-J. Duan*, R.-Q. Wang*, Photoinduced anisotropic edge states and signatures of nonequilibrium spin polarization in silicene nanoribbons, *Phys. Lett. A* **384,** 126429 (2020)

26. **J.-Y. Li**, R.-Q. Wang*, M.-X. Deng*, M. Yang, In-plane magnetization effect on current-induced spin-orbit torque in a ferromagnet/topological insulator bilayer with hexagonal warping, *Phys. Rev. B* **99,** 155139 (2019)

27. Y. Guo*, **J.-Y. Li**, M. Liu, Electromagnetically induced grating via coherently driven four-level atoms in a N-type configuration, *Int. J. Theor. Phys.* **54**, 868 (2015)

Peer review activities

• Reviewer for *Phys. Rev. Lett.*, *Phys. Rev. B*, and *npj Spintronics*.